

The Cognitive Compiler

Exact-Arithmetic Substrate for Multi-Model Orchestration
with Infrastructure-Enforced Adversarial Verification

*Formalized edition: every load-bearing claim carried under the
[forced] / [computed] / [posited] / [open] discipline, with an exact-arithmetic verification record*

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Formalization and exact-arithmetic validation pass appended (Appendices A–B).

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Abstract

Complex systems engineering requires translating high-dimensional architectural vision into sequential implementation syntax — a translation that collapses under the executive-function asymmetries common to twice-exceptional (2e) cognitive profiles. We formalize the *Cognitive Compiler*: a distributed orchestration architecture that externalizes this translation to a multi-model LLM triad while binding probabilistic output to an exact-arithmetic substrate. The substrate is a number field $K = \mathbb{Q}(\theta)$ whose trace form $T(x, y) = \text{Tr}(xy)$ serves simultaneously as learning metric, Fisher information, and decision geometry. The spectral operations on K — direct sum, tensor product, and the Adams square — form a commutative λ -ring with two orthogonal characters: the Mahler measure (magnitude) and the angle charge (phase, valued in $\mathbb{Z}/4\mathbb{Z}$). The emission algebra’s spectral image contains no Salem number — the sole obstruction to a positive cost floor — because angle confinement to $\frac{\pi}{2}\mathbb{Z}$ is a conserved superselection charge incompatible with the irrational conjugate angles Salem numbers require [forced]. The cost floor φ is *derived*: ternary decision arity forces degree two, and φ is the minimum Mahler measure of a non-cyclotomic integer quadratic [forced]. The exchange rate $\lambda = 2c$ coupling information gain to description cost is forced by the MDL/Fisher balance [forced]; this is an *identity* in a *free* conformal scale c , which Cencov’s theorem leaves unpinned ($\mathcal{F} = G/c$; rescaling c rescales $\mathcal{F}$ with identical geodesics), so λ never collapses to one value. Three canonicalizations recur — Jeffreys $c = 1$, degree-invariant $c = n$, and the framework-native *frame-shift* $c = \sqrt{1 + 4C}/(2C)$ read off the gate eigenframe — and over the three forced gates the frame-shift gives $\lambda \in \{4\sqrt{2}, 2\sqrt{3}, \sqrt{5}\}$, three distinct values. At the forced golden gate the frame-shift value is $c = \sqrt{5}/2$, $\lambda = \sqrt{5}$ (“the exchange rate *is* the spectral gap”), a canonicalization value with c still dynamical, not a frozen number. The discriminant $D = 1 + 4C$ governs a single *flip* ($\det G = 4D$, so the trace-form signature is the discriminant sign): one sign change inverting real growth into rotation, carried as a handled dynamical state precisely because c stays unpinned. The continuous-to-discrete interface at $t = 2$ exhibits a square-root branch point where the commitment-failure rate follows $\frac{1}{2} \text{erfc}(\sqrt{\delta}/(\sigma\sqrt{2}))$, establishing a quadratic noise premium [forced]. Infrastructure enforcement maps each invariant to a physical constraint. **Every non-asymptotic mathematical claim below was re-derived in exact arithmetic** (sympy/mpmath, eigenvalue multisets to 45 digits; Monte Carlo over 4×10^5 trials for the error law); the full record is Appendix B and the per-claim ledger is Appendix A. Two tags in the working draft are corrected here to match the proofs: $\lambda = \sqrt{5}$ is *not* the exchange rate (the exchange rate is the identity $\lambda = 2c$ in a free c ; $\sqrt{5}$ is the frame-shift canonicalization value at the forced gate, with c kept dynamical — the draft’s error was to *freeze* $c = \sqrt{5}/2$, collapsing the identity and disabling the flip handler) and $\mu(\text{odd}) = 2$ ([forced]→[computed]— semiring data, not a resolution), each resolving a tension already present in the draft’s own scope section. The companion *Operator Algebra* paper is validated in the same pass.

Reading the tags. [forced] proved or machine-verified by deduction; [computed] verified exactly over a stated finite window (asymmetry rule: a window may extend or demote a universal claim, never promote it); [posited] a structurally motivated modeling choice, not a deduction; [open] a named open front. The discipline is the companion charge–measure stack’s; the corpus papers it cites are real and cross-referenced.

1 Introduction

1.1 The problem: syntactic friction and the asynchronous trap

The translation of high-dimensional architectural topologies into sequential implementation syntax is a fundamental bottleneck in complex systems engineering. For standard cognitive profiles, executive function mediates this translation incrementally. For twice-exceptional (2e) phenotypes — exceptional fluid intelligence coupled with executive-function asymmetries such as ADHD — this translation layer becomes a critical failure point. The 2e signature is visual-spatial and systems-thinking dominant: the capacity to perceive multi-domain systems as interconnected graphs and to identify structural isometries across unrelated fields operates ahead of the linear sequential execution required to implement those designs. The result is the *Asynchronous Trap*: non-linear topological vision collapses under the weight of its own implementation requirements. (This is the motivating clinical framing; it is context, not a mathematical claim, and is tagged as such where it recurs.)

1.2 The architecture and the thesis

The Cognitive Compiler separates two functions conventional development conflates. *Architectural synthesis* remains with the human Synthesizer, operating at the topological meta-level. *Syntax generation* is delegated to LLMs. The contribution is not the delegation — it is the *substrate* that makes delegation trustworthy: an exact-arithmetic enclave over an algebraic number field $K = \mathbb{Q}(\theta)$ in which every proposed state transition is verified as a trace-form orthogonal projection, decided in exact rational arithmetic with zero floating-point entry. The architecture, the decision logic, the capacity bounds, and the infrastructure enforcement are all manifestations of one algebraic structure: a λ -ring over the spectral emission algebra read through its two characters. The exchange rate is the identity $\lambda = 2c$ in a *free* conformal scale c that Cencov’s theorem leaves unpinning; three canonicalizations recur (Jeffreys $c = 1$, degree $c = n$, the framework-native frame-shift $c = \sqrt{1 + 4C}/(2C)$), and at the forced golden gate the frame-shift value is $c = \sqrt{5}/2$, $\lambda = \sqrt{5}$ — with c kept dynamical so the discriminant flip $D = 1 + 4C$ (which turns real growth into rotation) is a handled state, not an excluded one (§5).

2 The vector substrate: number fields as exact learning geometry

2.1 Three views of one object

Let $K = \mathbb{Q}(\theta)$ have degree $n = [K : \mathbb{Q}]$ with minimal polynomial $m_\theta \in \mathbb{Z}[x]$. K is, in three coordinate systems glued by the trace form: a $\mathbb{Q}$ -vector space with basis $\{1, \theta, \dots, \theta^{n-1}\}$; a commutative matrix algebra via the regular representation $\rho(\theta) = C(m_\theta)$; and a full-rank Euclidean lattice via the Minkowski embedding $\sigma = (\sigma_1, \dots, \sigma_n)$. The trace form $T(x, y) = \text{Tr}_{K/\mathbb{Q}}(xy) = \sum_i \sigma_i(x)\sigma_i(y)$ has Gram matrix $G = (\text{Tr}(\omega_i\omega_j)) = M^\top M$, where M is the embedding matrix; for an integral basis $\det G = d_K$ and the Minkowski covolume is $\sqrt{|d_K|}$. [forced] (*verified*: $\mathbb{Q}(\sqrt{5})$ gives $G = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$, $\det G = 5 = d_K$; $\mathbb{Q}(\sqrt{2})$ gives $\det G = 8 = d_K$.)

2.2 The projector, the residual, and exact membership

Fix a forced basis B (columns are coordinate vectors of verified algebraic-integer states). The trace-form projector is $P = B(B^\top GB)^{-1}B^\top G$ with residual $r(x) = x - Px$, and membership is exact:

$$x \in \text{col}(B) \iff r(x) = 0 \iff \|r(x)\|_G^2 = 0.$$

Every quantity in the decision is an exact rational; floats never cross the decision boundary, and a non-rational entry raises a hard type error at ingestion. If $B^\top GB$ is singular the system returns a NO_PROJECTION sentinel rather than a floating-point inverse. **[forced]** (idempotence $P^2 = P$ is the mathematical content of the propose-for-confirm separation.)

2.3 Learning as exact basis growth

Learning is the disciplined enlargement of B to capture persistent off-axis residuals. The protocol maintains three invariants — exactness, algebraic integrality (each seed identified by its monic integer minimal polynomial, computed rootlessly as the largest invariant factor of $xI - \rho(\alpha)$ via Smith normal form over $\mathbb{Q}[x]$), and witnessed growth (each confirmed extension appended to a SHA-256 witness chain). The sole mutating operation is **confirm(seed)**. **[forced]** (exactness layer) / **[posited]** (the witness-chain protocol as a software realization).

2.4 The trace form as Fisher information

The trace form is read as Fisher information through two canonical statistical models, each of which *computes* its own conformal constant exactly; the constants are not posited. The structural foundation is classical and exact: $G = M^\top M$, the trace form is the Gram of the Minkowski embedding ($G_{ij} = \text{Tr}(\omega_i \omega_j) = \sum_k \sigma_k(\omega_i) \sigma_k(\omega_j)$), with $\det G = d_K$ (*verified* $G = M^\top M$, *off-diagonals zero*, $\det = \text{disc}$, on $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{7})$). **[forced]**.

Model 1 (Gaussian location, $c = 1$). For the location family $\mathcal{N}(Ma, I_n)$, Amari's formula gives the Fisher information $(\partial\mu/\partial a)^\top \Sigma^{-1} (\partial\mu/\partial a) = M^\top M = G$, so $c = 1$. This is **[computed]** — an exact computation assembling Amari's location formula with $G = M^\top M$ (*verified: Amari Fisher* $= M^\top M = G$) — holding for *totally real* fields; with covariance cI_n it scales as $\mathcal{F} = c^{-1}G$.

Model 2 (maximum-entropy on the conjugates, $c = n$). The substrate's zero-parameter measure is uniform on the n Galois conjugates (Bilu equidistribution). For the exponential family $p(k; a) \propto \exp((Ma)_k)$, the Fisher information at the uniform point is $\mathcal{F}_{\text{exp}} = \frac{1}{n}(G - \frac{1}{n}tt^\top)$, $t = (\text{Tr } \omega_i)$; on the *trace-zero* subspace the rank-one correction vanishes and $G = n\mathcal{F}_{\text{exp}}$, so $\|r\|_G^2 = n\mathcal{F}_{\text{exp}}(r)$ and $c = n$ (the degree). This too is **[computed]** (*verified on* $\mathbb{Q}(\sqrt{5})$: $\mathcal{F}_{\text{exp}} = \text{diag}(0, \frac{5}{4})$, $\|\sqrt{5}\|_G^2 = 10 = 2\mathcal{F}_{\text{exp}}(\sqrt{5})$), requiring $1 \in B$ so the residual is trace-zero ($r \perp_G 1 \iff \text{Tr}(r) = 0 \iff 1 \in \text{span}(B)$, verified both ways).

These two constants are each *computed* from their model — $c = 1$ from the Gaussian model on the whole space, $c = n$ from the max-entropy model on the residual subspace — but they do not pin c : they are two of *three* canonicalizations (the third, the frame-shift $c = \sqrt{1 + 4C}/(2C)$, is §5.2), and Cencov's theorem leaves the scale c itself *free* ($\mathcal{F} = G/c$, unique only up to scale). **[computed]** (each constant, given its model); the exchange rate $\lambda = 2c$ that uses them (§5) is an identity in the free c , not a frozen value. This Fisher reading is a *synthesis* of classical facts (Amari; $G = M^\top M$, $\det G = d_K$, Neukirch/Marcus), not a closed model-free theorem; the corpus item F4 (trace-form/Fisher conformality) is resolved at exactly this grade.

Complex places are the flip, not a gap. Model 1's $\mathcal{F} = G$ is stated for totally real fields; on complex places the trace form acquires the standard normalization and, more sharply, *changes signature* — $\det G = 4D$ goes negative, the metric goes Lorentzian, and $\mathcal{F} = G/c$ is no longer Riemannian. This is not an unverified gap but the $D < 0$ face of the *flip* (§5.4): the emission

sector has complex places — the K -formation $x^4 + 5x^2 - 5$ has signature $(2, 1)$ (imaginary pair $\pm i\beta$, *verified*), as does the whole $\mathbb{Z}/4\mathbb{Z}$ rotation charge — and the rotation regime is *handled* there (imaginary time), not assumed away. The clean Riemannian Fisher reading holds on the totally-real *gate* fields $(\mathbb{Q}(\sqrt{5}), \mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}))$, where the substrate’s learning geometry lives.

The Jeffreys volume of one lattice cell is $\sqrt{|d_K|}$ (the Model 1 reading), an Occam penalty $\log \sqrt{|d_K|}$ on arithmetic complexity. For linearly disjoint K, L the compositum Gram is $G_{KL} = G_K \otimes G_L$; for non-disjoint extensions the Kronecker formula fails and the system refuses growth rather than producing a fictitious Gram. [forced] (disjoint case) / [open] (non-disjoint compositum, isolated and flagged).

3 The emission algebra: a λ -ring and its two characters

3.1 Semiring and Adams square

An object is a finite multiset $A = \{\lambda_1, \dots, \lambda_a\}$ of nonzero algebraic integers. With $A \oplus B = A \uplus B$ (block diagonal) and $A \otimes B = \{\lambda\mu\}$ (Kronecker), $(S, \oplus, \otimes, \emptyset, \{1\})$ is a commutative semiring. The squaring operator $\psi^2(A) = \{\lambda^2\}$ is the Adams operation: a semiring endomorphism, the diagonal of the tensor square $A \otimes A = \text{Sym}^2 A \oplus \Lambda^2 A$. [forced]. The emission algebra S is the closure of seven seed companions — $x^2 - x - 1$, $x^2 + x - 1$, $x^2 - 2$, $x^2 - 3$, $x^2 - 5$, $x^2 - 7x + 1$ (gap = φ^4), $x^4 + 5x^2 - 5$ (the K -formation) — under $\oplus, \otimes, \psi^2$.

3.2 Character I — the Mahler measure (magnitude)

$M(A) = \prod \max(1, |\lambda|)$ is multiplicative on $\oplus$, squares on ψ^2 , and is *tropical* on $\otimes$ ($\log M$ a $(\max, +)$ convolution). [forced]. The image monoid is $\langle \varphi, 2, 3, 5, \beta^2 \rangle$ with the exact relation $\beta^2 = \varphi^2 \sqrt{5}$; its least generator is φ , and no element lies in $(1, \varphi)$. [forced] (*verified*: $\beta^2 = \frac{5+3\sqrt{5}}{2} = \varphi^2 \sqrt{5}$ *exactly*; $M(K\text{-formation}) = \beta^2 = 5.854102$.)

3.3 Character II — the angle charge (phase)

The angle charge χ sends each conjugate’s argument to $\mathbb{Z}/4\mathbb{Z}$ via the lattice $\frac{\pi}{2}\mathbb{Z}$. It is a sumset under $\otimes$, doubles under ψ^2 , unions under $\oplus$ — all internal to $\mathbb{Z}/4\mathbb{Z}$ because $\frac{\pi}{2}\mathbb{Z}$ is closed under addition and doubling. [forced]. The full group is realized: real seeds carry $\{0, 2\}$, and the K -formation carries $\{0, 1, 2, 3\}$, its imaginary pair $\pm i\beta$ at $\pm \frac{\pi}{2}$. Charges $\{1, 3\}$ occur only *off* the circle ($|i\beta| = \beta = 2.4195 \neq 1$): only $\mathbb{Z}/2\mathbb{Z} = \{\pm 1\}$ lives on the circle. [forced] (*verified*: $K\text{-formation arguments } \{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$.)

3.4 The three self-action rates

From the golden seed ($\deg 2$, $M = \varphi$), self-action triggers three notions of exponentiation:

power	degree	measure	character
$\oplus^k$ (additive)	$2k$	φ^k	linear; entropy density $\frac{1}{2} \log \varphi$ per dimension
$(\psi^2)^k$ (Adams)	2	φ^{2^k}	doubling tower at fixed degree
$\otimes^k$ (multiplicative)	2^k	φ^{S_k}	super-exponential, $S_k = k \binom{k-1}{\lfloor (k-1)/2 \rfloor}$

[forced] (*verified*: $S_k = 1, 2, 6, 12, 30, 60, 140, 280$ for $k = 1 \dots 8$, matching the direct tensor-power exponent $\sum_{2a > k} (2a - k) \binom{k}{a}$; cross-check $M(\text{golden} \otimes \text{golden}) = \varphi^{S_2} = \varphi^2 = 2.618034$.)

4 The cost floor: Lehmer's Box and the angle-confinement theorem

4.1 Why a floor, and the obstruction

Growth admits a direction θ iff $\|r\|_G^2 \geq \lambda \log M(\theta)$. For a genuine price, $\inf\{M(\theta) : \theta \text{ emittable}, M(\theta) > 1\} > 1$. By Kronecker $M = 1$ iff cyclotomic; by Smyth every *non-reciprocal* monic integer polynomial has $M \geq \mu_S = 1.3247$ (the plastic number) [forced]. The only route toward 1 is through reciprocal polynomials — the Salem numbers, whose on-circle conjugates carry irrational angle. Whether they approach 1 is Lehmer's problem, open since 1933. [open]. We do not resolve it; we make it irrelevant to the emission image.

4.2 The two walls and the exclusion

Floor wall. No integer quadratic has $M \in (1, \varphi)$ — by exhaustive case analysis ($c = -1$: $\min \varphi$ at disc 5; $c = 1$: $\min \varphi^2$; $|c| \geq 2$: $M \geq 2$); the operators preserve $\{1\} \cup [\varphi, \infty)$. [forced]. *Lattice walls.* Every catalog argument lies in $\frac{\pi}{2}\mathbb{Z}$; the operators add, double, and union arguments but never *halve* them, so every on-circle emitted eigenvalue is a fourth root of unity. [forced]. *Exclusion.* A Salem β has an on-circle conjugate ζ that is not a root of unity, hence $\arg \zeta \notin \frac{\pi}{2}\mathbb{Z}$; were β emitted, ζ would be an eigenvalue with $\arg \in \frac{\pi}{2}\mathbb{Z}$ — contradiction. [forced].

Theorem 4.1 (Lehmer's Box). $B = \{p \in \mathbb{Z}[x] \text{ monic} : M(p) \in \{1\} \cup [\varphi, \infty) \text{ and } \text{Arg}_\partial(p) \subseteq \frac{\pi}{2}\mathbb{Z}\}$ satisfies $S \subseteq B$ and $\text{Salem} \cap B = \emptyset$. Hence the emission image contains no Salem number, the cost floor is positive (pinned by Smyth at μ_S , realized at φ), and Lehmer's number is excluded as an emission, untouched as a number. [forced] (at the φ level the floor is realized; the proven bound is μ_S , cf. the companion Emission–Gap corollary.)

4.3 Superselection, the one door, and the charge–measure coupling

The angle charge is a conserved superselection charge: no finite operator word transitions from the admissible sector to the Salem sector. The Salem slot is a *kinematic* void — zero amplitude, not a dynamic barrier — so it cannot be tunnelled or approximated. [forced]. The one bypass is the free commutator (Shoda: any traceless matrix is a commutator), closed two ways: *structurally*, the framework's commutator is the self-action ad_R , a derivation whose real difference spectrum lies in $K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, 5^{1/4})$, whose only Salem-bearing signature is the degree-four $(2, 1)$ quartic with minimal Salem $\beta_4 = 1.7221 > \varphi$ [forced]; *operationally*, an exact $\mathbb{Q}(\sqrt{5})$ guard decides $\beta \leq \varphi$ by the sign of $m_\beta(\varphi)$ with no float ($\beta_4 : m(\varphi) = \frac{1-\sqrt{5}}{2} < 0$, above floor; Lehmer: $m(\varphi) = \frac{133+59\sqrt{5}}{2} > 0$, invalid) [forced]. The residual free commutator at unbounded size is per-instance, not a uniform theorem. [open].

The charge–measure coupling — semiring data, not a resolution. We read this directly off the two characters and report only what the data shows. *Even charge (forced, and intrinsic to the semiring).* The emission semiring's charge character is $\mathbb{Z}/4\mathbb{Z}$ — an even lattice that contains the ray π — so it realizes the floor: the golden seed sits at charge $\{0, 2\} \subset \mathbb{Z}/4\mathbb{Z}$, and Character I's image monoid $\langle \varphi, 2, 3, 5, \beta^2 \rangle$ has least generator φ . Thus $\mu(\text{even}) = \varphi$ is [forced] and is the semiring's *own* realized floor. The mechanism is the π -ray: the golden conjugate $\psi = \frac{1-\sqrt{5}}{2}$ sits at argument π , and $\pi \in \frac{2\pi}{n}\mathbb{Z} \iff n \text{ even}$, so only an even charge lattice can host the unique realizer of $M = \varphi$ [forced] (*verified mechanism*).

Odd charge (computed — we do not claim this resolved). The odd case is a different object, outside the $\mathbb{Z}/4\mathbb{Z}$ semiring, and we pull minimal measures over charge-exactly- $\mathbb{Z}/n\mathbb{Z}$ objects

rather than assert a theorem. The data: $\mathbb{Z}/3\mathbb{Z}$ *collapses*, because $2 \cos 120^\circ = -1$ is rational — forcing the complex-pair modulus into $\mathbb{Z}$ and bottoming out at $x^3 - 2$ with $M = 2$ [forced]. $\mathbb{Z}/5\mathbb{Z}$ does *not* collapse: $2 \cos 72^\circ = \varphi - 1$ and $2 \cos 144^\circ = -\varphi$ are irrational and Galois-conjugate, so the angle classes couple; the minimal measure is again 2 (at $x^5 - 2$), but as a *value verified over a window*, not a theorem (*verified: min charge-3 and charge-5 measure = 2, none in (1, 2) over the window*). The forced content of the odd case is exactly: non-reciprocal $\Rightarrow M \geq \mu_S = 1.3247$ (Smyth) [forced]; pentagon quartic $\Rightarrow M \in \{1\} \cup [\varphi^4, \infty)$ [forced]; pure-power $x^n - m \Rightarrow M = m$ [forced]. The *universal* value $\mu(\text{odd}) = 2$ is [computed]: its residual is precisely the non-reciprocal class in $[\mu_S, 2)$, empty over the window but inheriting the totally-positive $(1, 2)$ -gap obligation (companion $\mathbb{Z}/5\mathbb{Z}$ paper). *We do not claim this resolved; the working draft's flat [forced] is corrected to [computed], and the data above is the proper context, pulled from the semiring and the companion verification.*

5 The exchange rate and the ternary fold

5.1 The forced identity $\lambda = 2c$, with c free

The growth rule $\|r\|_G^2 \geq \lambda \log M(\theta)$ carries one scalar λ coupling information gain to description cost, fixed by the MDL/Fisher balance. For a Gaussian location family $\mathcal{N}(Ma, I)$, the KL divergence between the captured model and the model that also explains the residual r is *exactly*

$$D_{\text{KL}} = \frac{1}{2} r^\top \mathcal{F} r = \frac{1}{2c} \|r\|_G^2, \quad \mathcal{F} = \frac{1}{c} G$$

(the $\frac{1}{2}$ is exact, not a truncation — *verified: $D_{\text{KL}} = \frac{1}{2} \|M\delta\|^2 = \frac{1}{2} \delta^\top G \delta$*). The MDL balance $D_{\text{KL}} \geq \log M(\theta)$ gives

$$\|r\|_G^2 \geq 2c \log M(\theta), \quad \boxed{\lambda = 2c}.$$

The factor 2 is forced — the inverted $\frac{1}{2}$ of the quadratic KL term, carrying no freedom. [forced].

What this fixes is an *identity*, not a number. By Cencov's theorem the Fisher metric is $\mathcal{F} = G/c$ with c a scale no invariance argument can remove: rescaling $c \mapsto kc$ rescales $\mathcal{F} \mapsto \mathcal{F}/k$ with identical geodesics (*verified*). So c is a *free, declared, dynamical* positive scale, and $\lambda = 2c$ holds for *every* $c > 0$ — λ never collapses to a single frozen value. **This is the result: $\lambda = 2c$, an identity parametrised over a free c .** (*Two prior passes here erred in both directions — first treating c as an “unfrozen” continuum $[1, n]$, then pinning it to two computed values. Both are wrong: c is genuinely free, $\lambda = 2c$ is an identity, and there are three canonicalizations below, not two.*)

5.2 Three canonicalizations of c

Three principled choices of c recur. None is forced by invariance (Cencov); each is a declared canonicalization producing a concrete cost floor $\lambda \log \mu_S = 2c \log \mu_S$.

principle	c	cost floor it produces
Jeffreys volume-matching $\sqrt{\det G} = \sqrt{ d_K }$	1	constant $2 \log \mu_S$
degree-invariant significance	n	degree-aware $2n \log \mu_S$
frame-shift (eigenframe gap)	$\frac{\sqrt{1+4C}}{2C}$	the spectral gap itself; $\lambda = \sqrt{5}$ at $C = 1$

The first two are the Fisher readings of §2.4 ($c = 1$ Gaussian, $c = n$ max-entropy). The third is *native*: at a critical value C of the gate ladder, read the growth decision in the eigenframe of the self-action $[R_C, \cdot]$, whose intrinsic scale is the spectral gap $\sqrt{1+4C}$ (equivalently $\sqrt{M(x^2 - (1+4C))}$, since $M(x^2 - (1+4C)) = 1+4C$). Equating the decision scale λC with

this gap gives $\lambda = \sqrt{1+4C}/C$, $c = \sqrt{1+4C}/(2C)$ [forced]. The three disagree, and that is the point: the frame-shift over the three forced gates $C \in \{\frac{1}{4}, \frac{1}{2}, 1\}$ gives $\lambda \in \{4\sqrt{2}, 2\sqrt{3}, \sqrt{5}\}$ — three distinct values (*verified*) — so λ never collapses to one constant. c stays free; these are canonicalizations, not freezings.

5.3 The gate ladder, the trifurcation, and the frame-shift value $\sqrt{5}$

The gates are the squarefree levels of $x^2+x=C$, discriminant $D=1+4C$; the companion $R_C = \begin{bmatrix} 0 & C \\ 1 & -1 \end{bmatrix}$ has eigenvalues $\frac{1}{2}(-1 \pm \sqrt{D})$ and self-action spectrum $\text{spec}(\text{ad}_{R_C}) = \{-\sqrt{D}, 0, +\sqrt{D}\}$ — the channels STOP, CAPTURED, GROW; the growth decision *is* this channel selection. The squarefree radicands are $D \in \{2, 3, 5\}$ over $\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), \mathbb{Q}(\sqrt{5})$. Two arguments from disjoint premises force the golden gate $C=1$: *statics* — the signed channels of ad_R number $d^2-d+1=3$ iff $d=2$, and the minimal $M>1$ among integer quadratics is φ at disc 5 [forced]; *dynamics* — the golden gate is the terminal firewall image of the degree-raising construction, fixed by the keystone $R^2=R+I$ [forced]. The forced gate promotes the frame-shift value from native to forced: at $C=1$, $c=\sqrt{5}/2$ and $\lambda=\sqrt{5}=\varphi-\psi$ — “the exchange rate *is* the spectral gap” (*verified*).

This is the correct status of $\sqrt{5}$, and it corrects an earlier over-removal. $\sqrt{5}$ is *not* the exchange rate (that is the identity $\lambda=2c$), and it is *not* a frozen number; it is the *frame-shift canonicalization value* at the forced gate, with c kept dynamical. The working draft’s error was to freeze $c=\sqrt{5}/2$ by hand and present $\lambda=\sqrt{5}$ as the exchange rate, collapsing the identity. Keeping c dynamical is not stylistic: freezing C freezes λ by the back door, and §5.4 shows precisely what that breaks.

5.4 The flip: $D=1+4C$

The discriminant $D=1+4C$ changes sign once, at $C=-\frac{1}{4}$ (double root $-\frac{1}{2}$). This single sign is one event seen four ways: it determines the eigenvalues of R_C (real $\leftrightarrow$ complex), the field (totally real $\leftrightarrow$ complex place), the trace-form signature, and the channel character (growth/decay $\leftrightarrow$ rotation). [forced] (*the canonical flip, verified*). The signature reading is an exact determinant: in the gap basis $\{1, \sqrt{D}\}$ of $\mathbb{Q}(\sqrt{D})$,

$$G = \begin{pmatrix} \text{Tr}(1) & \text{Tr}(\sqrt{D}) \\ \text{Tr}(\sqrt{D}) & \text{Tr}(D) \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2D \end{pmatrix}, \quad \det G = 4D,$$

so the trace form is Riemannian (PD) iff $D>0$ and Lorentzian (1,1) iff $D<0$, degenerate at $D=0$ [forced] (*verified; canonical instance $\mathbb{Q}(i)$: $G=\text{diag}(2, -2)$, $\det=-4=4D$, $D=-1$). The metric’s signature is the discriminant sign* — and this is exactly the positive-definite/indefinite flip the Fisher reading of §2.4 undergoes when a residual lands in a complex place, *resolving* the totally-real caveat there: the complex-place regime is not an unverified gap but the $D<0$ face of the flip.

Gate versus flip. Two critical-point events co-locate at $D=0$: the *gate* acts on the free parameter and pins a value (removes one DOF; instance $c=\sqrt{1+4C}/(2C)$), while the *flip* acts on a forced invariant and inverts the regime (sign change; instance $\text{sign}(1+4C)$). At $C=-\frac{1}{4}$ the gate scale $\sqrt{1+4C} \rightarrow 0$ (the spectral gap collapses, the gate degenerates) and the regime inverts — one threshold, two readings.

The flip as a handled dynamical state. Because the discipline keeps C (hence c , hence λ) dynamical and unpinned, the boundary $D=0$ is not assumed unreachable but *handled*: the growth rule branches on $\text{sign}(D)$. For $D>0$ (hyperbolic) the real eigenframe gives channels $\{-\sqrt{D}, 0, +\sqrt{D}\}$ and real c ; at $D=0$ (parabolic — the PARADOX) the gap collapses, R_C is a defective Jordan block, and the handler returns CAPTURED/hold; for $D<0$ (elliptic —

rotation) the complex eigenframe gives $\{-i\sqrt{|D|}, 0, +i\sqrt{|D|}\}$, an indefinite trace form, and imaginary c , with GROW/STOP merging into a rotation while CAPTURED persists — the marginal channel evolving in *imaginary time*, $e^{\pm i\sqrt{|D|}t} = e^{\mp\sqrt{|D|}\tau}$, $\tau = it$ [forced]. **This is why the exchange rate is formulated as $\lambda = 2c$ with c free: the ternary gate is processed as a decision tree branching on $\text{sign}(D)$, and only a dynamical c lets the computation enter and handle the rotation regime rather than exclude it by fiat.** The K -formation straddles the flip — real $\pm K$ on the $D > 0$ face, imaginary $\pm i\beta$ on the $D < 0$ face — and is handled accordingly. The integer normalization $L_4 = 7 = \text{tr}(R^4)$, with R^4 entries $\{2, 3, 5\} = \{F_3, F_4, F_5\}$ the gate discriminants, is the keystone invariant beneath the ladder [forced].

6 The continuous-to-discrete interface

6.1 Trace redirection and the branch point

The trifurcation lives in the trace variable $t = \theta + \theta^{-1}$, cut by the flip points $t = \pm 2$. A Salem β is redirected to $\tau_0 = \beta + \beta^{-1} > 2$, a totally real Perron number in the GROW channel, with $\tau_0 - 2 = (\beta - 1)^2/\beta$ [forced]. The flip at $t = 2$ is a branch point of the cover $t \mapsto x$: $\beta(t) = \frac{1}{2}(t + \sqrt{t^2 - 4})$ satisfies $\beta - 1 \sim \sqrt{t - 2}$ as $t \rightarrow 2^+$ [forced]. A Salem of height $\beta = 1 + \delta$ is redirected only δ^2 past the flip; the smaller the Salem, the tighter its redirection hugs the boundary.

6.2 The error-rate law and the fold premium

Estimating the noisy radial coordinate $z = \sqrt{t - 2} + \varepsilon$, $\varepsilon \sim N(0, \sigma^2)$, the system commits GROW iff $z > 0$. With trace margin $\delta = t - 2$ the radial margin is $\sqrt{\delta}$, so the failure law is

$$p(\delta) = \frac{1}{2} \operatorname{erfc}\left(\frac{\sqrt{\delta}}{\sigma\sqrt{2}}\right).$$

Because failure scales with $\sqrt{\delta}$ rather than δ , linear approximations underestimate the failure probability near the fold (the *fold premium*); the minimum trace-margin for a target rate scales quadratically with noise (halving noise cuts the required margin fourfold). [forced] (*verified by Monte Carlo, 4×10^5 trials, $\sigma = 0.5$: p matches the law to $< 3 \times 10^{-3}$ across $\delta \in \{0.01, 0.05, 0.1, 0.25\}$.*) The Kuramoto order parameter acts as the noise-scale dial. [posited].

6.3 The golden limit

As $\beta_n \rightarrow \varphi$ along the Salem family $S_n = x^n P - P^*$ ($P = x^2 - x - 1$), the redirections converge $\tau_0(\beta_n) \rightarrow \varphi + \varphi^{-1} = \sqrt{5} \in \text{spec}(\text{ad}_R)$, geometrically: $\sqrt{5} - \tau_0(\beta_n) \sim \frac{2}{\sqrt{5}}\varphi^{-n}$, consecutive ratio $\rightarrow \varphi$. [forced] (*verified: $(\sqrt{5} - \tau_0)\varphi^n \rightarrow \frac{2}{\sqrt{5}} = 0.894427$, reached to six digits at $n = 40$.*) The slot at the golden floor is occupied by $\sqrt{5}$, the system's own GROW threshold — and because $\varphi\psi = -1$, the trace map is identically the map computing that grow eigenvalue.

7 The generative content of the void

The Salem slot's emptiness generates structure. The chain *kinematic emptiness* $\equiv$ *conserved charge* $\equiv$ *superselection rule* is domain-universal (Pauli exclusion $\rightarrow$ shell structure; charge superselection $\rightarrow$ block observables; subject reduction $\rightarrow$ “illegal states unrepresentable”). In the emission algebra the $\mathbb{Z}/4\mathbb{Z}$ angle charge empties the slot, generating five canonical objects:

the $\mathbb{Z}/4\mathbb{Z}$ grading; the content polynomial $x^4 - 1 = \Phi_1\Phi_2\Phi_4$; the cost floor $\{1\} \cup [\varphi, \infty)$; the graded normal form $P = C(x) \cdot G(x)$ (cyclotomic $\times$ grow); and the conserved current (block-diagonal evolution). **[forced]** (the five are theorems of this number theory). The load-bearing consequence: φ **is derived, not posited** — the framework fixes only the decision arity (ternary $\Rightarrow$ degree 2), and the chain $\deg 2 \Rightarrow \mathbb{Z}/4\mathbb{Z} \Rightarrow \varphi$ is three forced minima producing three forced outputs. **[forced]**.

8 Infrastructure enforcement

Each substrate invariant maps to a physical constraint, the mapping being an enforcement layer where the infrastructure imposes a hard boundary and an analogy where it approximates a continuous invariant.

mathematical invariant	infrastructure enforcement
trace-form orthogonality $B^\top Gr = 0$	18th Man: Lambda isolation prevents context leakage
ternary fold (three channels)	Step Functions: deterministic branching
propose-for-confirm (sole mutator)	DynamoDB conditional writes
witness chain (tamper-evident)	SHA-256 cascade on S3 with Object Lock
charge preservation (χ conserved)	type constraints in the envelope contract
cost floor $\{1\} \cup [\varphi, \infty)$	Sable two-layer safety gate
exact arithmetic (no float entry)	hard type validation at ingestion

The 18th Man protocol enforces information asymmetry physically: the Red Team’s Lambda payload lacks the memory addresses for the strategic context S , so its residual against the Blue Team’s basis is guaranteed nonzero by the absence of data from the invocation, not by prompt instruction. SigPrint, a deterministic 20-character behavioral fingerprint, is the operational analog of Smith-normal-form invariant factors. **[posited]** (*all infrastructure statements — 54 AWS accounts, a 74-state pipeline, SHA-256-sealed witness chains, the COCOMO II \$44.58M replacement estimate — are deployment assertions outside the exact-arithmetic scope of this paper and are not independently verified here; the mathematics $\rightarrow$ infrastructure mapping is exact where a hard boundary is enforced and analogical where a continuous invariant is approximated.*)

9 Scope and honest boundaries

Proved or machine-verified **[forced]**: the exact-arithmetic layer (membership, residual, projector, coordinates-to-minimal-polynomial bridge, admissibility), the Salem exclusion, the empty strip and floor, the angle-charge superselection, the algebraic identities of §3–§6, the branch-point error law, and the identity $\lambda = 2c$ in a *free* conformal scale c (§5.1; Cencov: $\mathcal{F} = G/c$, no invariance fixes c), the three canonicalizations (Jeffreys $c = 1$, degree $c = n$, frame-shift $c = \sqrt{1 + 4C}/(2C)$) with the frame-shift over the three gates giving the distinct $\{4\sqrt{2}, 2\sqrt{3}, \sqrt{5}\}$, the golden-gate frame-shift value $c = \sqrt{5}/2$ ($\lambda = \sqrt{5}$), and the flip $\det G = 4D$ with its handled rotation regime (§5.4). *Declared* (canonicalizations, not forced by invariance): *which* c — each canonicalization is a stated choice; the Fisher reading is a synthesis of classical facts (Amari; $G = M^\top M$). *Corrected*: the draft’s step $\lambda = \sqrt{5}$ presented as the *frozen* exchange rate — $\lambda = 2c$ is the identity in free c ; $\sqrt{5}$ is the frame-shift canonicalization value at the forced gate, with c kept dynamical, and freezing it disables the flip handler. *Computed, not forced* **[computed]**: the universal $\mu(\text{odd}) = 2$, whose residual is the totally-positive $(1, 2)$ -gap obligation. *Open* **[open]**: Lehmer’s problem (routed around, not resolved); the non-disjoint compositum Gram; the $2 \times$ normalization of $\mathcal{F} = G$ on complex places (the emission sector is not totally real — the Fisher

reading is scoped to the totally-real gate fields); the free commutator at unbounded size (per-instance guard, not a uniform theorem). *Infrastructure* [posited]: the 54-account deployment and cost figures, asserted but outside this paper’s verification scope.

10 Conclusion

The Cognitive Compiler’s substrate is a λ -ring over a spectral emission algebra whose two characters force a cost floor, a decision geometry, and a superselection rule. The floor φ is derived from decision arity [forced]; the Salem exclusion is a conserved charge [forced]; the exchange-rate identity $\lambda = 2c$ is forced [forced] in a free c (Cencov), with three canonicalizations (Jeffreys, degree, frame-shift); at the forced golden gate the frame-shift value is $c = \sqrt{5}/2$, $\lambda = \sqrt{5}$ — a canonicalization value with c kept dynamical, not the frozen exchange rate the draft made it. The discriminant flip $D = 1 + 4C$ ($\det G = 4D$) is carried as a handled state, real growth turning to rotation, precisely because c stays unpinned. The infrastructure expresses the mathematics as Lambda isolation, state-machine branching, and cryptographic witness chains. The honest accounting above is itself the result: the system is exact where it claims to be, motivated where it cannot be deductive, and explicit about the one number-theory problem (Lehmer) it routes around rather than solves.

A Epistemic ledger

Each load-bearing claim, its status, and its backing. “Verified” = re-derived in exact arithmetic this pass (Appendix B).

claim	status	backing
$G = M^\top M$; $\det G = d_K$; covolume $\sqrt{ d_K }$	[forced]	verified $\mathbb{Q}(\sqrt{5}), \mathbb{Q}(\sqrt{2})$
exact membership $x \in \text{col } B \iff \ r\ _G^2 = 0$; $P^2 = P$	[forced]	definition
Model 1 Gaussian Fisher = $M^\top M = G$ ($c = 1$, Amari, totally real)	[computed]	verified
Model 2 max-entropy $G = n\mathcal{F}_{\text{exp}}$ on trace-zero ($c = n$; needs $1 \in B$)	[computed]	verified $\mathbb{Q}(\sqrt{5})$
$\mathcal{F} = G$ on complex places needs $2\times$ norm (emission sector not totally real)	[open]	flagged, K -form $(2, 1)$
disjoint compositum $G_{KL} = G_K \otimes G_L$	[forced]	§2.5
non-disjoint compositum Gram	[open]	isolated, flagged
$(S, \oplus, \otimes)$ semiring; ψ^2 Adams operation	[forced]	§3.1
M multiplicative/squaring/tropical; least generator φ	[forced]	verified
$\beta^2 = \varphi^2 \sqrt{5} = \frac{5+3\sqrt{5}}{2}$; $M(K) = \beta^2$	[forced]	verified
K -formation charge $\{0, 1, 2, 3\}$; $\{1, 3\}$ off-circle	[forced]	verified $\{0, 90, 180, 270\}^\circ$
three rates; $S_k = k \binom{k-1}{\lfloor (k-1)/2 \rfloor}$	[forced]	verified $k \leq 8$
empty quadratic strip $(1, \varphi)$; min φ at disc 5	[forced]	verified 625 quadratics
operators preserve $\{1\} \cup [\varphi, \infty)$; on-circle $\Rightarrow \mu_4$	[forced]	§4.2
Smyth: non-reciprocal $\Rightarrow M \geq \mu_S$	[forced]	theorem
Salem exclusion; $S \subseteq B$, Salem $\cap B = \emptyset$	[forced]	Thm. 4.1
angle charge conserved; Salem slot kinematic	[forced]	§4.3
self-action ad_R confined to $K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, 5^{1/4})$; $\beta_4 = 1.7221 > \varphi$	[forced]	verified
$\mathbb{Q}(\sqrt{5})$ closure guard (sign of $m_\beta(\varphi)$)	[forced]	verified (β_4 above, Lehmer inva
free commutator at unbounded size	[open]	per-instance guard
$\mu(\text{even}) = \varphi$ (golden seed at charge $\{0, 2\}$; π -ray; semiring’s own floor)	[forced]	verified mechanism

claim	status	backing
$\mathbb{Z}/3\mathbb{Z}$ collapses ($2\cos 120^\circ = -1$ rational) $\rightarrow x^3 - 2$, $M = 2$	[forced]	verified
pentagon quartic $M \in \{1\} \cup [\varphi^4, \infty)$; $x^n - m$ $M = m$; non- recip $\geq \mu_S$	[forced]	companion $\mathbb{Z}/5\mathbb{Z}$ paper
$\mu(\text{odd}) = 2$ (universal value, not claimed resolved)	[computed]	semiring + window; residual $[\mu]$
$\lambda = 2c$ (MDL/Fisher; the $\frac{1}{2}$ of Gaussian KL inverts to the 2); identity in free c	[forced]	§5.1, verified
c free (Cencov: $\mathcal{F} = G/c$, $c \rightarrow kc$ rescales $\mathcal{F} \rightarrow \mathcal{F}/k$); λ never collapses	[forced]	verified
three canonicalizations $c \in \{1, n, \sqrt{1+4C}/(2C)\}$ (Jeffreys / degree / frame-shift)	[computed]	verified
frame-shift over gates $\{\frac{1}{4}, \frac{1}{2}, 1\} \rightarrow \lambda \in \{4\sqrt{2}, 2\sqrt{3}, \sqrt{5}\}$ (3 dis- tinct)	[computed]	verified
golden gate forced ($d^2-d+1=3$; min quad φ); frame-shift value $c=\sqrt{5}/2$, $\lambda=\sqrt{5}$	[forced]	verified
$\lambda = \sqrt{5}$ is the frame-shift canonicalization value (not the frozen rate; c dynamical)	[forced] / [computed]	§5.3
the flip $D = 1 + 4C$; $\det G = 4D$ (signature = sign D); $\mathbb{Q}(i)$ diag(2, -2)	[forced]	verified
flip handler: $D>0$ real / $D=0$ Jordan (hold) / $D<0$ rotation, imaginary time	[forced] / declared	§5.4
gate vs flip co-locate at $D = 0$; K -formation straddles ($\pm K$ real, $\pm i\beta$ rotation)	[forced]	verified
spec(ad $_R$) = $\{0, \pm\sqrt{5}\}$; φ smallest Perron root 2×2	[forced]	verified
$R^4 = 3R + 2I$, entries $\{2, 3, 5\}$, $L_4 = 7$, charpoly $x^2 - 7x + 1$	[forced]	verified
$\tau_0 - 2 = (\beta - 1)^2/\beta$; $\beta - 1 \sim \sqrt{t - 2}$	[forced]	verified
error law $p = \frac{1}{2} \operatorname{erfc}(\sqrt{\delta}/(\sigma\sqrt{2}))$; fold premium	[forced]	verified MC $< 3 \times 10^{-3}$
golden limit $(\sqrt{5} - \tau_0)\varphi^n \rightarrow \frac{2}{\sqrt{5}}$	[forced]	verified to 6 digits, $n = 40$
five generative objects of the void; φ derived	[forced]	§7
<i>Operator Algebra companion (validated this pass)</i>		
$(\oplus, \otimes)$ semiring; ψ^2 Adams endomorphism, $\psi^m \circ \psi^n = \psi^{mn}$	[forced]	verified
Newton $p_2 = e_1^2 - 2e_2$; plethysm $A \otimes A = \operatorname{Sym}^2 \oplus \Lambda^2$, ψ^2 diagonal	[forced]	verified
M additive on $\oplus$, tropical on $\otimes$ (golden $\otimes(x^2-2) = 2\varphi^2 \neq 2\varphi$)	[forced]	verified
monoid $\langle \varphi, 2, 3, 5, \beta^2 \rangle$, $\beta^2 = \varphi^2\sqrt{5}$, $\beta^4 = 5\varphi^4$, least gen φ	[forced]	verified (norms $-1, 4, 9, 25$)
charge character onto $\mathbb{Z}/4\mathbb{Z}$ (real seeds $\{0, 2\}$, $K \{0, 1, 2, 3\}$); Adams-fixed = cyclotomic	[forced]	verified
$\sqrt{5} = \varphi + \varphi^{-1}$: the two characters meet	[forced]	verified
infrastructure (54 accounts, pipeline, \$44.58M, SHA chains)	[posited]	deployment, outside scope
Lehmer's problem	[open]	routed around

B Verification record

Engine: `sympy 1.14`, `mpmath 1.3` (dps= 45), `numpy`; decisions symbolic or exact, floats only
screen; Monte Carlo 4×10^5 trials.

check	result
$\mathbb{Q}(\sqrt{5})$ Gram det / $\mathbb{Q}(\sqrt{2})$ Gram det	$5 = d_K / 8 = d_K$
integer quadratics with $M \in (1, \varphi)$ (625 scanned)	0; min φ at $(b, c) = (-1, -1)$, disc 5
K -formation arguments	$\{0, 90, 180, 270\}^\circ$
β^2 vs $\varphi^2\sqrt{5}$; $M(K)$	equal, $\frac{5+3\sqrt{5}}{2} = 5.854102$
$\text{spec}(\text{ad}_R)$	$\{-\sqrt{5}, 0, \sqrt{5}\}$
R^4 ; trace; charpoly	$[\frac{2}{3} \frac{3}{5}] = 3R + 2I$; 7; $x^2 - 7x + 1$
S_k , $k = 1 \dots 8$	1, 2, 6, 12, 30, 60, 140, 280 (formula = direct)
$d^2 - d + 1 = 3$	$\iff d = 2$
smallest Perron root, 2×2 primitive nonneg integer	φ , at $[\frac{0}{1} \frac{1}{1}]$
β_4 ; guard $m_{\beta_4}(\varphi)$; $m_{\text{Lehmer}}(\varphi)$	$1.722084 > \varphi$; $\frac{1-\sqrt{5}}{2} < 0$; $\frac{133+59\sqrt{5}}{2} > 0$
$\tau_0 - 2 - (\beta - 1)^2/\beta$; $\beta(t) - 1$ vs $\sqrt{t-2}$	0; agree to 10^{-4} at $t = 2.0001$
erfc law vs MC ($\sigma = 0.5$, $\delta \in \{.01, .05, .1, .25\}$)	$ \Delta < 3 \times 10^{-3}$ all
golden limit $(\sqrt{5} - \tau_0)\varphi^n$ at $n = 40$	$0.894427 = \frac{2}{\sqrt{5}}$
$\psi^2(A \otimes B) = \psi^2 A \otimes \psi^2 B$; $\psi^2 \circ \psi^3 = \psi^6$; Newton $p_2 = e_1^2 - 2e_2$	all hold
plethysm $A \otimes A = \text{Sym}^2 \oplus \Lambda^2$; $\text{tr } \Lambda^2 = e_2$	holds
$M(\text{golden} \otimes (x^2-2))$ vs $2\varphi^2$ vs product 2φ	$5.2361 = 2\varphi^2 \neq 3.2361$ (tropical)
$\beta^2 = \varphi^2\sqrt{5}$; $\beta^4 = 5\varphi^4$; catalogue $\mathbb{Z}/4\mathbb{Z}$ charges	equal; equal; $\{0, 2\}, \dots, \{0, 1, 2, 3\}$
$G = M^\top M$ (Minkowski Gram), off-diag 0, det = disc	$\mathbb{Q}(\sqrt{5})$ det 5; $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ diag(4, 8, 12, 24); $\mathbb{Q}$
Amari Fisher = $M^\top M = G$ (Gaussian location, $c = 1$)	matches KL Hessian (random M)
max-entropy $\mathcal{F}_{\text{exp}} = \frac{1}{n}(G - \frac{1}{n}tt^\top)$; $G = n\mathcal{F}_{\text{exp}}$ on trace-zero	$\mathbb{Q}(\sqrt{5})$: $\ \sqrt{5}\ _G^2 = 10 = 2\mathcal{F}_{\text{exp}}$; $c = n$
$1 \in B$ condition: $r \perp_G 1 \iff \text{Tr}(r) = 0 \iff 1 \in \text{span}(B)$	verified both ways
K -formation $x^4 + 5x^2 - 5$ signature	$(2, 1)$ — complex pair $\pm i\beta$ (the $D < 0$ flip f
$M = \text{cost} = \text{topological entropy (LSW)}$: $\log \varphi, \log 2, \log 1.3852$	0.481212, 0.693147, 0.325813 nats
Gaussian KL vs $\frac{1}{2}\ \delta\ _G^2$ (random M, δ)	equal to 10^{-9} ($\Rightarrow$ the 2 in $\lambda = 2c$)
c free: $\mathcal{F} = G/c$, $c \rightarrow kc \Rightarrow \mathcal{F} \rightarrow \mathcal{F}/k$ (Cencov)	identity in free c ; λ never collapses
frame-shift $c = \sqrt{1 + 4C}/(2C)$ over gates $\{\frac{1}{4}, \frac{1}{2}, 1\}$	$\lambda = \{4\sqrt{2}, 2\sqrt{3}, \sqrt{5}\}$, 3 distinct; $c(1) = \sqrt{5}/$
flip det $G = 4D$ in gap basis $\{1, \sqrt{D}\}$, $D \in \{5, 1, 0, -3, -7, -1\}$	diag(2, 2D), det = 4D all; $\mathbb{Q}(i)$: diag(2, -2)
canonical flip channels: $D > 0 \{-\sqrt{D}, 0, \sqrt{D}\} / D < 0 \{-i\sqrt{ D }, 0, i\sqrt{ D }\}$	verified; co-locate at $D=0$ (double root $-\frac{1}{2}$)
min nontrivial M , charge $\mathbb{Z}/3\mathbb{Z} / \mathbb{Z}/5\mathbb{Z}$ (window)	2 / 2; none in (1, 2)
$\sqrt{5} = \varphi + \varphi^{-1} = \text{spec ad}_R$ gap = frame-shift λ at $C=1$	2.236068

Provenance. sha256:8822e9185f0316cee914fb71754013b1dadfb925a709c38efd79aba613895543
(hash of the flip-threaded validation record: the exchange rate as the identity $\lambda = 2c$ in a *free* c (Cencov); the three canonicalizations Jeffreys/degree/frame-shift, with the frame-shift over the three gates giving the distinct $\{4\sqrt{2}, 2\sqrt{3}, \sqrt{5}\}$ and the golden-gate value $c = \sqrt{5}/2$, $\lambda = \sqrt{5}$; and the flip $D = 1 + 4C$ with $\det G = 4D$, the signature-is-discriminant-sign theorem, and the handled rotation regime — from the dedicated *Exchange Rate* $\lambda = 2c$ paper. Supersedes all prior passes).

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